1M1B AI FOR SUSTAINABILITY VIRTUAL INTERNSHIP

In Partnership with IBM SkillsBuild & AICTE

Anna-Setu

AI-Powered Campus Food Rescue Network

Final Solution Submission File

Pankhuri Govila

THE MANDATE

Contextual SDG Alignment

Anna-Setu is engineered against two interlocking United Nations Sustainable Development Goals.

12

PRIMARY TARGET

SDG 12: Responsible Consumption & Production

Objective Framework: Halving macro-level municipal and consumer-level food losses throughout standard operational pipelines by 2030.

2

SECONDARY TARGET

SDG 2: Zero Hunger

Objective Framework: Safeguarding high-quality nutritional continuity and access loops across peripheral communities using safely managed surplus channels.

THE PROBLEM

Targeted Institutional Gaps

> ***How might we use AI to accurately forecast daily meal demand and intelligently match unavoidable surplus food with verified local shelters, so that campus dining ecosystems and food-insecure communities can become more sustainable?***

CORE INSTITUTIONAL FRICTIONS IDENTIFIED



Rigid Menu Cycles

Fixed, recurring operational menus lack structural input sensitivity tools — blind to real-time demand signals.



Default Overproduction

Kitchen crews default to fixed overproduction margins to avoid customer friction, with no data to justify the buffer.



No Compliance Layer

Absence of standard compliance layers for local food drops blocks safe, confident redistribution logistics.

THE SOLUTION

Deep-Dive System Solution

MODULE 1

The Advisor

Collects calendar variations, weather conditions, and attendance trends to deliver explicit kitchen preparation guides - with stated confidence metrics - before every shift starts.

MODULE 2

The Pipeline

Automatically formats data inputs into clean, compliance-checked notifications the moment verified surplus limits are detected - routing them to the nearest matched shelter.

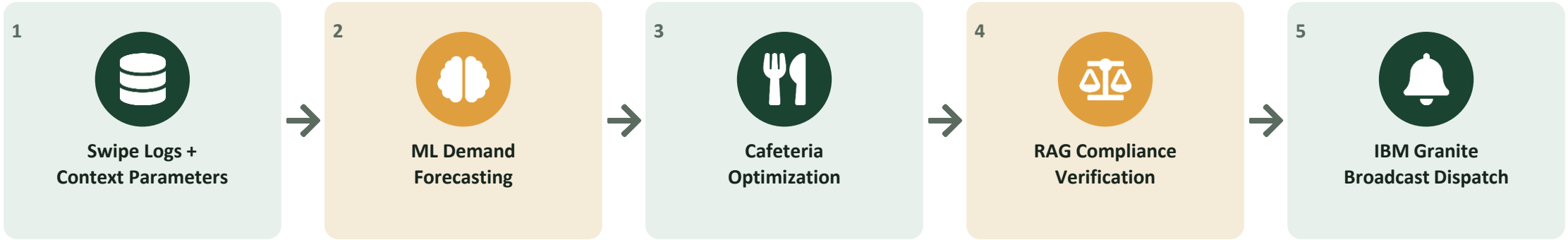
TARGET USERS

- **Dining Hall Managers**
- **Local NGO Shelter Operators**
- **Sustainability Oversight Committees**

THE ARCHITECTURE

Strategic System Architecture

DATA FLOW MECHANICS



This pipeline is fully implemented as a working prototype.

See it live in action → "Anna Setu Demo App Screenshot" on the next slide.

THE PROMPTING LAYER

System Prompt Engineering Matrix

SYSTEM DIRECTIVE

“You are an automated, secure communications dispatch layer processing details for Anna-Setu. Map out concise parameters for text notification alerts without adding unverified details.”

STRUCTURED OUTPUT MATRIX

Yields highly clear, standardized text messages mapping:

- **Localized surplus items**
- **Verified volumes on hand**
- **Clear timing deadlines**
- **Explicit geographic coordinates**

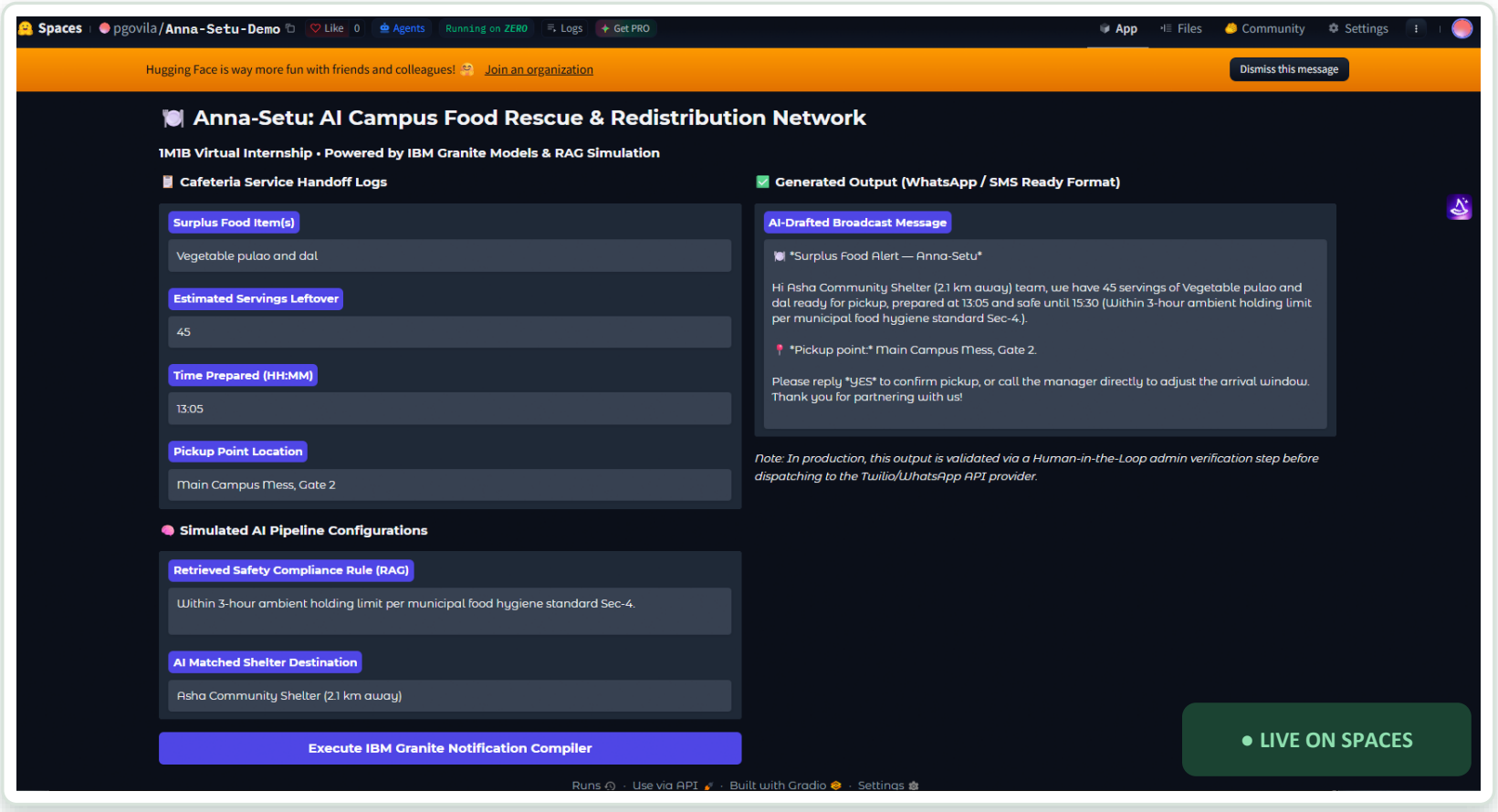
This exact prompt logic runs inside a live, deployed prototype.

See it live in action → “Anna Setu Demo App Screenshot” on the next slide.

APP SCREENSHOT

Anna Setu Demo Application Screenshot

A fully functional Gradio prototype simulating the end-to-end Anna-Setu pipeline — deployed live on Hugging Face Spaces.



WHAT THIS DEMO SHOWS

Cafeteria Handoff Logs

Structured input: surplus item, servings, prep time, pickup point.

RAG Compliance Retrieval

Live-retrieved safety rule: 3-hour ambient holding limit, Sec-4.

AI-Matched Shelter

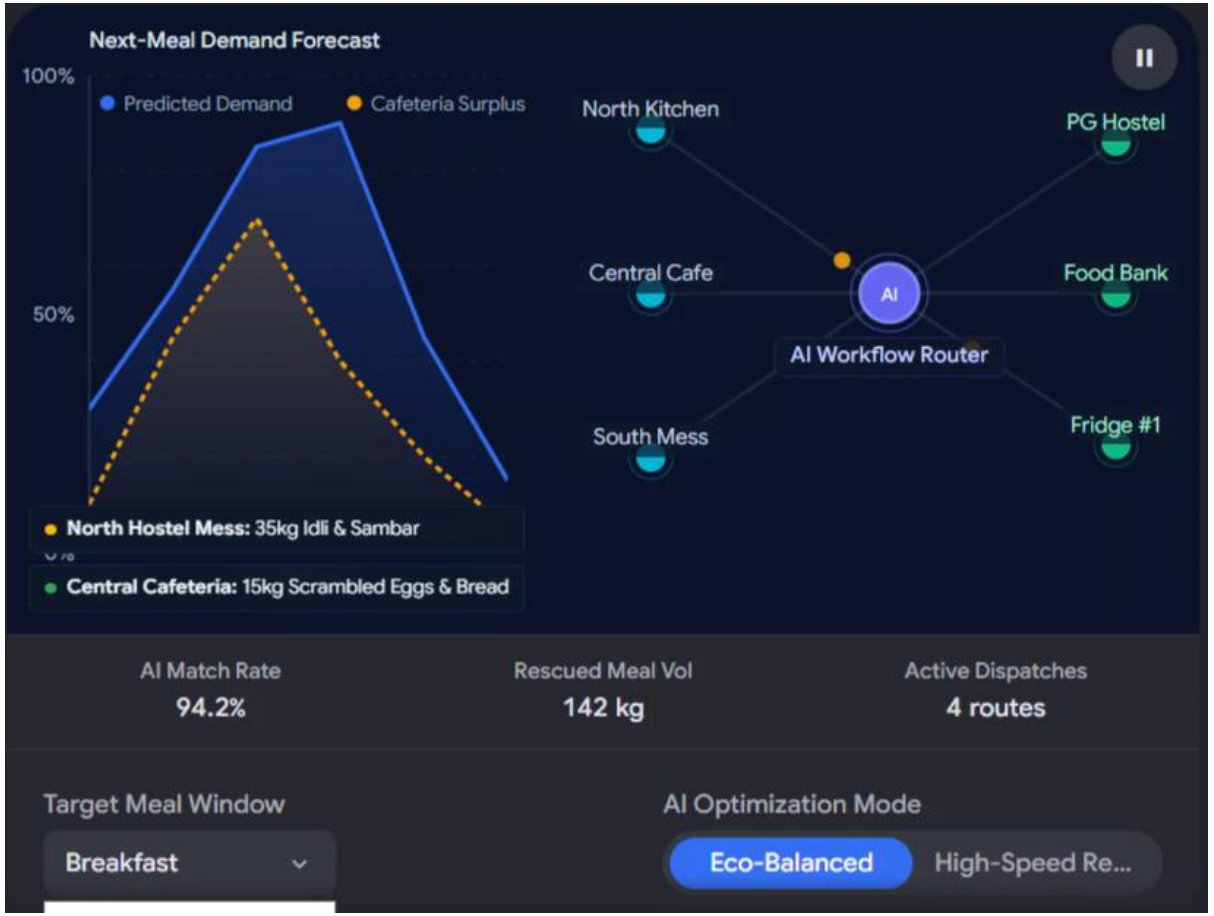
Nearest verified partner auto-selected — Asha Community Shelter.

Granite Broadcast Output

Ready-to-send WhatsApp/SMS alert, drafted end-to-end by the model.

DASHBOARD DEMO

Visual Architecture and Workflow AI Dashboard



THE IMPACT

Expected Project Impacts

15–25%

Economic Vectors

Target reduction in raw food ingredient overproduction losses, lowering municipal waste-hauling costs.

↓ CH_4

Environmental Vectors

Prevents leftover food batches from ending up in local landfills, lowering methane gas production variables.

24/7

Social Vectors

Establishes predictable, safe access channels to direct unused campus food resources directly to community centers.

THE SAFEGUARDS

Ethically Governed Operations

Mandatory Responsible AI Considerations

FAIRNESS

Demand forecasting metrics rely entirely on aggregated, de-identified cohort traffic logs. The system treats all dining complexes equally without tracking individual student habits or preferences.

TRANSPARENCY

The dashboard delivers plain-language explainability matrices alongside every projection (e.g., “12% turnout reduction predicted due to mid-semester exam scheduling”), enabling kitchen staff to confidently audit or override recommendations.

ETHICS

All system actions are checked by a strict rules engine. The tool automatically blocks any redistribution alert if calculations show a food batch has exceeded local ambient storage timelines.

PRIVACY

Meal validation inputs are stripped of student identifiers before being processed, ensuring no individual health profiles or eating habits are stored or shared.

DEPLOYMENT

Repository & System Deployment Anchors

Live artifacts confirming Anna-Setu functions as a working, deployable system.



GITHUB PROJECT FILES

https://github.com/pgovila/Anna-Setu-Project



INTERACTIVE LIVE SIMULATION URL

huggingface.co/spaces/pgovila/Anna-Setu-Demo